

Modelling, Dynamics and Kinematics

Sample Exam

Formula sheet

$$x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \Leftrightarrow \tilde{x} = \begin{pmatrix} 0 & -x_3 & x_2 \\ x_3 & 0 & -x_1 \\ -x_2 & x_1 & 0 \end{pmatrix} \quad (1)$$

$$r = \frac{\omega \times v}{\|\omega\|^2}; \quad \lambda = \frac{\omega^\top v}{\|\omega\|^2} \quad (2)$$

$$\text{Ad}_{H_a^b} = \begin{pmatrix} R_a^b & 0 \\ \tilde{p}_a^b R_a^b & R_a^b \end{pmatrix}; \quad \text{ad}_{T_a^{c,b}} = \begin{pmatrix} \tilde{\omega}_a^{c,b} & 0 \\ \tilde{v}_a^{c,b} & \tilde{\omega}_a^{c,b} \end{pmatrix} \quad (3)$$

$$e^A := I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \dots \quad (4)$$

$$\theta \tilde{\omega} \in \mathfrak{so}(3), \quad \|\hat{\omega}\| = 1 \Rightarrow e^{\theta \tilde{\omega}} = I + \tilde{\omega} \sin \theta + \tilde{\omega}^2 (1 - \cos \theta) \quad (5)$$

$$e^{\tilde{T}} = \begin{pmatrix} e^{\tilde{\omega}} & \frac{1}{\|\omega\|^2} ((I - e^{\tilde{\omega}})(\omega \times v) + (\omega^\top v) \omega) \\ 0 & 1 \end{pmatrix} \quad (6)$$

$$\hat{T}_{\text{rot}} = \begin{pmatrix} \hat{\omega} \\ r \times \hat{\omega} \end{pmatrix}; \quad \hat{T}_{\text{tr}} = \begin{pmatrix} 0 \\ \hat{v} \end{pmatrix}; \quad \hat{T}_{\text{scr}} = \begin{pmatrix} \hat{\omega} \\ r \times \hat{\omega} + \lambda \hat{\omega} \end{pmatrix} \quad (7)$$

$$\frac{d}{dt} \text{Ad}_{H_b^a} = \text{Ad}_{H_b^a} \text{ad}_{T_b^{b,a}} \quad (8)$$

$$\text{ad}_{T_{\bullet}^{b,\bullet}} = \text{Ad}_{H_a^b} \text{ad}_{T_{\bullet}^{a,\bullet}} \text{Ad}_{H_b^a} \quad (9)$$

$$(\dot{\mathcal{P}}^i)^\top = \text{ad}_{T_i^{i,0}}^\top (\mathcal{P}^i)^\top + (W^i)^\top \quad (10)$$

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = \tau^\top \quad \left(G(q) = \frac{\partial V}{\partial q} \right) \quad (11)$$

Part I (Test)

Read carefully. Wrong answers are penalised, i.e. the normalised Part I score is calculated as

$$\text{Score} = \frac{\# \text{ Correct answers} - \# \text{ Wrong answers}}{\# \text{ Questions}},$$

while the unanswered questions are not counted as either correct or wrong. In the questions 1 to 8, circle T if the statement is true and F if it is false.

1. Consider an n -dimensional real vector space V^n .

- T F Every possible basis of V^n contains infinitely many elements.
- T F The dimension of the Lie Algebra of V^n is $\frac{n(n-1)}{2}$.
- T F Any vector $v \in V^n$ possesses a unique representation of n components $v^i \in \mathbb{R}$ with respect to a basis $e_i \in V^n$, $i \in \{1, \dots, n\}$, namely the one such that $v = \sum_{i=1}^n v^i e_i$ holds.
- T F A map transforming components of vectors in V^n between two different bases is always invertible and has $\mathbb{R}^n$ as both domain and co-domain.
- T F A change of basis is always a linear map.
- T F The map $\phi: V^n \rightarrow \mathbb{R}$ defined as $v \mapsto \phi(v) := 0$ for all $v \in V^n$ is not a covector.
- T F The map ϕ defined above is the null vector of the covector space $(V^n)^*$.

2. Consider the Lie group $\text{SO}(3)$.

- T F It is an infinite-dimensional manifold.
- T F It is a finite-dimensional manifold.
- T F It is both a manifold and an Abelian group (meaning that $\forall g, h \in \text{SO}(3)$, $g \bullet h = h \bullet g$, where $\bullet$ denotes the group operation).
- T F A basis for its Lie algebra contains three elements.
- T F It is both a manifold and a non-Abelian group.
- T F An element of $\text{SO}(3)$ is represented by a 3×3 matrix.

3. A homogeneous matrix $H_a^b \in \text{SE}(3)$:

- T F represents the rigid motion from frame Ψ_a to frame Ψ_b expressed in Ψ_a ;
- T F represents the rigid motion from frame Ψ_b to frame Ψ_a expressed in Ψ_b ;
- T F represents the rigid motion from frame Ψ_b to frame Ψ_a expressed in Ψ_a ;
- T F has the inverse given by $H^{-1} = H^\top$;
- T F is an element of a three-dimensional Lie group.
- T F $\dot{H}_a^b$ being left-translated to the identity is a twist expressed in Ψ_b .
- T F Homogeneous matrices can be composed using the chain rule since they are $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ -tensors.

4. Consider a 1-DoF kinematic pair connecting body a to body b . The joint state is described by the variable q .
- T F Torque τ produces the mechanical power $\tau(\dot{q})$ (covector τ applied to vector $\dot{q}$).
- T F The unit twist $\hat{T}_b^{a,a}$ is always constant.
- T F The kinematic relation is given by $H_b^a(q) = \exp(\hat{T}_b^{a,a} q) H_b^a(0)$.
- T F The exponential map $\exp: \mathfrak{se}(3) \rightarrow \text{SE}(3)$ is surjective but not injective since there exist $T_1, T_2 \in \mathfrak{se}(3)$ with $T_1 \neq T_2$ such that $e^{T_1} = e^{T_2}$.
- T F $H_a^b = -H_b^a$
5. The geometric Jacobian:
- T F does not depend on the configuration of the robot;
- T F contains the unit twists that characterise the kinematic pairs of the robot expressed in the base frame Ψ_0 ;
- T F has always a non-zero determinant if the robot is redundant;
- T F may lose rank in some configurations;
- T F is a $\binom{0}{2}$ -tensor;
- T F represents a linear transformation from $T_q\mathcal{Q}$ to $\mathfrak{se}(3)$.
6. The Brockett's formula:
- T F solves the inverse kinematics problem for serial-chain manipulators;
- T F is computationally convenient since it provides a closed form solution which depends only on the current joint positions q , the unit twists of the joints and the homogeneous matrix of the final link in the reference configuration ($q = 0$);
- T F solves the direct kinematic problem for serial-chain manipulators;
- T F produces a 4×4 matrix as a result, which represents an element of $\text{SE}(3)$;
- T F produces a matrix as a result, whose last row contains only zeroes.
7. Consider the dynamics of a rigid body $(\dot{\mathcal{P}}^i)^\top = \text{ad}_{T_i^{i,0}}^\top(\mathcal{P}^i)^\top + (W^i)^\top$, also referred to as Lie-Poisson equation.
- T F The superscript i refers to a frame which is attached to the rigid body, since the equation is in the body-fixed frame.
- T F The first three elements of the vector $\text{ad}_{T_i^{i,0}}^\top(\mathcal{P}^i)^\top$ represent rotational gyroscopic effects and take the form $(J\omega_i^{i,0}) \times \omega_i^{i,0}$ (where J is a diagonal 3×3 matrix) if i refers to the principal inertia frame.
- T F In order for the equation to be numerically solved, one needs the information on the current configuration $H_i^0(t)$ of the rigid body.
- T F W^i is an external wrench acting on the rigid body.

- T F It is an equation on the space $\mathfrak{se}^*(3)$.
- T F It is valid only in an inertial frame, like Newton's law for a point mass $\dot{p} = f$.
8. The inertia (mass) matrix $M(q)$ in the Euler–Lagrange equations of a robot:
- T F is always a square matrix.
- T F is invertible only if the robot is not in a singular configuration.
- T F defines uniquely the so-called Coriolis matrix $C(q, \dot{q})$.
- T F is a $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ -tensor on the configuration manifold $\mathcal{Q}$.
- T F is a $\begin{pmatrix} 0 \\ 2 \end{pmatrix}$ -tensor since $\frac{1}{2}\dot{q}^\top M(q)\dot{q}$ is the scalar kinetic energy in the system.
- T F has zero determinant in some configurations, called singular configurations.

Part II (Open question)

Brockett's formula is used to calculate the end-effector configuration $H_n^0(q)$ for a serial robot, given the joint coordinates q . Derive the expression for Brockett's formula for a three-degree-of-freedom manipulator starting from the chain rule for H matrices and the kinematic pair relation:

$$H_i^j(q_i) = e^{\tilde{T}_i^{j,j} q_i} H_i^j(0).$$

Explain each step in your derivation. Keep all the indices.

Matrix exponential property $Be^AB^{-1} = e^{BAB^{-1}}$ might be helpful.

Comment on each variable, e.g. q , $\tilde{T}$, $H(0)$, in the final expression. *What does the variable represent? What space does it belong to?*

Part III (Exercise)

Consider the 4-DoF robot, the frames and distances represented in the figure. The robot consists of:

- A translational joint (q_1) with axis aligned in direction y of the base frame Ψ_0 ;
- A rotational joint (q_2) as in figure;
- A rotational joint (q_3) as in figure;
- A translational joint (q_4) allowing an extension of the vertical arm.

The point E is part of the end-effector and has constant coordinates $(0,0,1)^T$ in Ψ_4 , i.e. has homogeneous coordinates $E^4 = (0,0,1,1)^T$ in Ψ_4 . Make a choice for the reference configuration ($q_i = 0, i \in \{1,2,3,4\}$) considering that the distance A in the direction of the vertical arm is a fixed constant which cannot be shrunk (i.e. the vertical arm has length $A + q_4$ with $q_4 \geq 0$).

1. Give Brockett's equation for direct kinematics and all unit twists in reference configurations that you need to calculate $H_4^0(q)$. *Note: There is no need to write out the matrix exponentials but you must calculate the vector products that appear inside your expressions.* (1 Point)
2. Compute the geometric Jacobian $J(q)$ such that $T_4^{0,0} = J(q)\dot{q}$. *Note: You must calculate the vector products that appear inside your expressions.* (2 Points)
3. Compute the velocity of the point E in the inertial frame Ψ_0 as function of q and $\dot{q}$. (0.5 Points)
4. Suppose an external wrench is applied on the end-effector with coordinates in Ψ_4 given by $W^4 = (0 \ 0 \ 0 \ 0 \ 1 \ 0)$, i.e. the wrench is constant in the end-effector frame. This could be the case if e.g. a propeller mounted on the end-effector would produce a constant thrust. Compute the generalised forces, as function of q and $\dot{q}$, which are experienced by the joints as effect caused by this wrench. (0.5 Points)

